# The localisation of the poset model

a complement to *A model of univalence over a base category with an interval*

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**Abstract**

The poset instance of the model (`model.tex`, §8.1) is the presheaf model over finite non-empty posets. Following Sattler, it is localised at an operation $\mathsf{D}$ built from the inclusion of the linear posets and the strictly monotone maps (`~/POSET/poset-model.pdf`). We state the part of that construction which is proved there, in the setting of `model.tex`: $\mathsf{D}$ is a strict monad on types; if $A$ is fibrant then so is $\mathsf{D}A$; and the two maps $\mathsf{D}A \to \mathsf{D}^2A$ are connected by a path and are homotopy equivalences. The composition of $\mathsf{D}A$ is the face-uniform filling of `model.tex`, Theorem 8.3(2), first step, applied to the restriction of $A$ to the simplex category; no excluded middle is used. The passage to a left exact modality is the criterion of Coquand, Ruch and Sattler, which is quoted, not proved.

## 1 Linear posets and the restriction to $\Delta$

We work over the poset instance: $\mathcal{C}$ is the category of finite non-empty posets, $\Phi(X)$ the decidable sieves, Cl = Fib. The linear posets $[n] = \{0 < \cdots < n\}$ are objects of $\mathcal{C}$, and the simplex category $\Delta$ is the full subcategory on them; $\Delta_+ \subseteq \Delta$ has the strictly monotone maps, which are the composites of face maps. The simplicial instance of `model.tex` §8.2 is built on these same maps: its objects are the numbers $n$, its maps $[m] \to [n]$ the maps of $\mathcal{C}$.

**Definition 1.1** (restriction)**.** For a context $\Gamma$ over $\mathcal{C}$, $i^*\Gamma$ is the simplicial set $n \mapsto \Gamma([n])$; for a type $A$ over $\Gamma$, $i^*A$ is the type over $i^*\Gamma$ with $(i^*A)(n, \tau) = A([n], \tau)$. Restrictions are those of $\Gamma$ and $A$ along the maps of $\Delta$.

A line of $\Gamma$ at $X$ (`model.tex` §2.2), read at the generalised points $(f, j) : [W] \to X \times [1]$ with $[W]$ linear, gives for every $f : [n] \to X$ a line of $i^*\Gamma$ at $n$: $(g, j) \mapsto \gamma(fg, j)$. A partial line of $A$ over $\gamma$ on a sieve $\varphi$ of $X$ gives in the same way a partial line of $i^*A$ on the sieve $f^*\varphi$ of $[n]$. So a filling problem for $A$ at $X$ gives a filling problem for $i^*A$ at every $f : [n] \to X$, and this problem is formed from the data restricted along $f$ only.

## 2 Horn filling from composition

**Lemma 2.1.** *A fibration structure on $A$ over $\Gamma$ gives a horn filling operation on $i^*A$ over $i^*\Gamma$ (`model.tex`, Definition of horn filling), natural in $\Gamma$.*

*Proof.* Let $\tau \in \Gamma([m+1])$, $0 \le k \le m+1$, and a partial element of $i^*A$ on $\Lambda^k[m+1]$; it is given by its values $a_Y$ on the faces $Y$ of the horn, which are linear. If $k = 0$, let $q$ be the constant map 0 and $g_Y : Y \times [1] \to Y$, $g_Y(y, 0) = q(y)$, $g_Y(y, 1) = y$. The open box of $A$ at $[m+1]$ over the

constant line at $\tau$, with the sides $a_Y g_Y$ on the sieve generated by the faces of the horn and the lower lid $a_{\{0\}}\, q$, is composed from $e_0$ to $e_1$; the end is a filler, since $g_Y(y,1) = y$. If $k \geq 1$, the same with $q(x) = \max(x,k)$, which maps every face of the horn into itself and $[m+1]$ onto the proper face $\{x \geq k\}$, the roles of 0 and 1 exchanged, and the composition from $e_1$ to $e_0$. The construction uses only the data and the composition, so it is natural. □

This is Theorem 8.3(1) of `model.tex` with the retraction made explicit, here for a type over posets read on the linear posets.

## 3 The operation D

**Definition 3.1.** Let $A$ be a type over $\Gamma$. For $\rho \in \Gamma(X)$, $\mathsf{D}A(X,\rho)$ is the set of families $u(f) \in A([n], \rho f)$, for all $n$ and $f : [n] \to X$, with

$$u(f)\, g = u(fg) \qquad \text{for every strictly monotone } g : [m] \to [n].$$

Restriction along $k : X' \to X$ is $(uk)(f) = u(kf)$. For $\alpha : A \to B$, $(\mathsf{D}\alpha\, u)(f) = \alpha(u(f))$.

It suffices to require the condition for the face maps. The family and the condition are coded (a family indexed by $n$ and the maps $[n] \to X$, a proposition), so $\mathsf{D}A$ is a type. $\mathsf{D}$ commutes with substitution: for $s : \Delta \to \Gamma$, $\mathsf{D}(A[s])$ maps to $\mathsf{D}A$ by a cartesian map over $s$ whose fibre maps transport the values along the naturality of $s$, as for $\Pi$ (`model.tex`, Theorem 4.5).

**Definition 3.2.** $\eta_A : A \to \mathsf{D}A$, $(\eta_A a)(f) = af$; and $\mu_A : \mathsf{D}^2 A \to \mathsf{D}A$, $(\mu_A v)(f) = v(f)(\mathrm{id})$.

**Lemma 3.3** (strict monad)**.** *$\eta$ and $\mu$ are natural, $\mathsf{D}\alpha \circ \mu_A = \mu_B \circ \mathsf{D}^2\alpha$, and*

$$\mu_A \circ \eta_{\mathsf{D}A} = \mathrm{id}, \qquad \mu_A \circ \mathsf{D}\eta_A = \mathrm{id}, \qquad \mu_A \circ \mathsf{D}\mu_A = \mu_A \circ \mu_{\mathsf{D}A},$$

*as equalities of maps, that is of codes.*

*Proof.* Unfold the definitions (poset-model, Lemma 7.4). □

## 4 The composition of D$A$

**Theorem 4.1.** *A horn filling operation on $i^*A$ gives a fibration structure on $\mathsf{D}A$. Taken on the universal family, it is natural in $\Gamma$. With Lemma 2.1: if $A$ is fibrant, so is $\mathsf{D}A$.*

*Proof.* Let $\gamma$ be a line of $\Gamma$ at $X$, $u_0 \in \mathsf{D}A(\gamma(e))$, and $(\varphi, u)$ a partial line of $\mathsf{D}A$ ending at $u_0$. For $f : [n] \to X$ the data read along $f$ (Section 1) form a filling problem for $i^*A$ at $n$: the line $(g,j) \mapsto \gamma(fg, j)$, the end $u_0(f)$, and the partial line on $f^*\varphi$ whose value at $(g,j)$ is $u_{fg}$ at the corresponding chain. These values are natural only along the injective maps of $\Delta$; they are extended to a simplicial partial line by the Eilenberg–Zilber factorisation: factor $\langle g, j\rangle = m\sigma$ with $m$ an injective chain of the prism and $\sigma$ surjective, and take the value at $m$ restricted along $\sigma$. The factorisation is that of the composite point of the prism, not of a stage of the partial line: a degenerate restriction of a stage would otherwise be read through a non-injective map. This is natural along every map, since the factorisation is unique and $v(m)m' = v(mm')$ for $m'$ injective, and it commutes with restriction along face maps; only its values on chains are used below. Let $F(f)$ be its *face-uniform* filler (`model.tex`, proof of Theorem 8.3(2), first step: fill the shuffles $P_n, \ldots, P_0$ one after the other, each by one horn filling, and take the side when it contains the identity). The composite is $f \mapsto F(f)(e')$.

- It is in $\mathsf{D}A$: a face of the filler at $f$ is the filler at the face of $f$, since the face-uniform fillers are uniform along face maps, and faces suffice.
- It agrees with the sides: if $f$ lands in $\varphi$, the side of the problem contains the identity and the filler is that side.
- It is uniform along every $k : X' \to X$: the problem of the restricted data at $f$ is the problem of the given data at $kf$, so the two fillers are the same.

The horn filling operation is only used on the prisms $[n] \times \Delta[1]$, so it needs no uniformity, and the face-uniform fillers need no excluded middle.

Naturality is obtained as for Ext (`model.tex`, §6.2), by making the construction once on the universal family. $\mathsf{El}$ is fibrant (`model.tex`, §8.1), so $\mathsf{DEl}$ over $\mathsf{U}$ has the fibration just built. Every fibrant type of the poset model is $c^*\mathsf{El}$ for its code map $c : \Gamma \to \mathsf{U}$; its $\mathsf{D}$ carries the fibration of $\mathsf{DEl}$ moved along the cartesian map $\mathsf{D}(c^*\mathsf{El}) \to \mathsf{DEl}$ over $c$ (Section 3). For $s : \Delta \to \Gamma$ the cartesian maps compose, so the structure on $\mathsf{D}(A[s])$ is the move of that on $\mathsf{D}A$ by the composition law of moves (`model.tex`, Lemma 3.9); the fillers themselves are never moved. □

*Remark* 4.2. Only the uniformity along the face maps is used. This is why $\mathsf{D}A$ is fibrant without the normalised fillers, which require excluded middle and make fillers uniform along the degeneracies as well: the condition defining $\mathsf{D}A$ only concerns $\Delta_+$.

## 5 Paths and the two maps $\mathsf{D}A \to \mathsf{D}^2A$

A *homotopy* between maps $\alpha, \beta : A \to B$ over $\Gamma$ is an element of the path type $\mathrm{Path}_B(\alpha, \beta)$ over $\Gamma.A$ (`model.tex`, Theorem 4.5): at $(\rho, a)$, a line in $B$ over the constant line at $\rho$ from $\alpha(a)$ to $\beta(a)$, natural in $(\rho, a)$. We write $\alpha \simeq \beta$; a *homotopy equivalence* is a map with a two-sided homotopy inverse. Homotopies compose with maps on both sides.

**Lemma 5.1.** *If $\alpha \simeq \beta$ then $\mathsf{D}\alpha \simeq \mathsf{D}\beta$.*

*Proof.* For a generalised point $F = (f, h) : [n] \to X \times [1]$ put $H'(u)(F) = H(u(f))\langle \mathrm{id}, h\rangle$, where $\langle \mathrm{id}, h\rangle : [n] \to [n] \times [1]$ (poset-model, Lemma 9.2). □

**Lemma 5.2.** *$\eta_{\mathsf{D}A} \simeq \mathsf{D}\eta_A$, by a homotopy which uses no composition or filling of $A$.*

*Proof.* For $F = (f, h)$, $h$ cuts $[n] = A_0 + A_1$; a map $g : [m] \to [n]$ cuts $[m] = B_0 + B_1$, with $g = (g_0 + A_1)(B_0 + g_1)$. Put $v(F)(g) = u\big(f(g_0 + A_1)\big)(B_0 + g_1)$. Every condition on $v$ is an instance of one principle: $u(F)K$ is unchanged when $(F, K)$ is reparametrised along a strictly monotone map (poset-model, Lemma 9.3). □

**Corollary 5.3.** *$\mu_A \circ \mathsf{D}\eta_A = \mathrm{id}$ and $\mathrm{id} \simeq \mathsf{D}\eta_A \circ \mu_A$, with no fibrancy assumption. If $A$ is fibrant, $\mathsf{D}\eta_A$ and $\eta_{\mathsf{D}A}$ are homotopy equivalences.*

*Proof.* $\mu_A \circ \mathsf{D}\eta_A = \mathrm{id}$ by Lemma 3.3. By Lemma 5.2 and Lemma 5.1, $\mathsf{D}\eta_{\mathsf{D}A} \simeq \mathsf{D}^2\eta_A$; composing with $\mu_{\mathsf{D}A}$ on the left and using Lemma 3.3, $\mathrm{id} = \mu_{\mathsf{D}A} \circ \mathsf{D}\eta_{\mathsf{D}A} \simeq \mu_{\mathsf{D}A} \circ \mathsf{D}^2\eta_A = \mathsf{D}\eta_A \circ \mu_A$. For $\eta_{\mathsf{D}A}$: $\mu_A \circ \eta_{\mathsf{D}A} = \mathrm{id}$, and $\eta_{\mathsf{D}A} \circ \mu_A \simeq \mathsf{D}\eta_A \circ \mu_A$ by Lemma 5.2. When $A$ is fibrant so is $\mathsf{D}^2A$ (Theorem 4.1 twice), and homotopies into a fibrant type can be inverted and composed, by composition in the target; so $\eta_{\mathsf{D}A} \circ \mu_A \simeq \mathrm{id}$ and $\mathsf{D}\eta_A \circ \mu_A \simeq \mathrm{id}$. □

## 6 The universal property of $\eta$

A fibrant type $B$ is $\mathsf{D}$-*modal* when $\eta_B$ is a homotopy equivalence. By Corollary 5.3, $\mathsf{D}A$ is $\mathsf{D}$-modal for every fibrant $A$ (condition 1 of the criterion below).

**Proposition 6.1.** *Let $A$ be a type and $B$ a fibrant $\mathsf{D}$-modal type over $\Gamma$, with $p_B : \mathsf{D}B \to B$ and $p_B \circ \eta_B \simeq \mathrm{id}$. For $u : A \to B$ put $G(u) = p_B \circ \mathsf{D}u : \mathsf{D}A \to B$. Then $G(u) \circ \eta_A \simeq u$, and $G(f \circ \eta_A) \simeq f$ for every $f : \mathsf{D}A \to B$. So precomposition with $\eta_A$ is a bijection up to homotopy between the maps $\mathsf{D}A \to B$ and the maps $A \to B$.*

*Proof.* $G(u) \circ \eta_A = p_B \circ \mathsf{D}u \circ \eta_A = p_B \circ \eta_B \circ u \simeq u$, by the naturality of $\eta$ (strict). And $G(f \circ \eta_A) = p_B \circ \mathsf{D}f \circ \mathsf{D}\eta_A \simeq p_B \circ \mathsf{D}f \circ \eta_{\mathsf{D}A} = p_B \circ \eta_B \circ f \simeq f$, by Lemma 5.2 and the naturality of $\eta$. The homotopies are composed in the fibrant type $B$. □

This is the argument of [CRS], Theorem 1.10, (2)⇒(1), for maps over $\Gamma$ rather than for the internal function types.

## 7 What is quoted, and what may follow

With Theorem 4.1, Lemma 5.2, Corollary 5.3 and Lemma 3.3, the criterion of Coquand, Ruch and Sattler (poset-model, Theorem 10.1) makes $\mathsf{D}$ a left exact modality, and the $\mathsf{D}$-modal types form a constructive model of univalent type theory with higher inductive types. That criterion is not formalised. Two further results of poset-model use only the structures above and could be added: a proposition with a point over [0] has an element of $\mathsf{D}A$ (Proposition 11.4), and $\mathsf{D}A$ is trivially fibrant when spheres bound in $A$ (Theorem 12.3, the trivial case of Whitehead's principle).

## A Correspondence with the formalisation

Layer L9 of `seg/core`, over the poset instance; all statements of this note are formalised except those of the last section.

| statement | files |
|---|---|
| Section 1 | CLocRestr |
| Lemma 2.1 | CLocHorn |
| Definitions of $\mathsf{D}$, $\eta$, $\mu$, Lemma 3.3 | CLocD |
| Theorem 4.1 | CLocEZ (the factorisation), CLocCompV, CLocCompP, CLocComp; naturality on the universal family: CLocDCart, CLocDUniv |
| homotopies, Lemma 5.1 | CHtpyOps (L3: homotopies, whiskering, inverse and composition into a fibrant type), CLocHtpy |
| Lemma 5.2 | CLocPath, CLocPathEnd, CLocEtaH |
| Corollary 5.3 | CLocEq |
| Proposition 6.1 | CLocUniv (isModal, dModal; UP.PA.resA, UP.PBm.resB) |